Cell Imaging using Lateral Shearing Interferometer leading to cell characterization

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CERTIFICATE

This is to certify that **Daksh Patel** student of Applied Physics with **PRN:2017033800139383** has successfully completed the dissertation entitled **"Cell Imaging using Lateral Shearing Interferometer leading to cell characterization"** in the academic year 2021-2022 and submitted the work to department as per requirement of the degree of Masters of Science in Applied Physics under the guidance of **Dr. Arun Anand** in Department of Applied Physics, Faculty of Science and Technology, The Maharaja Sayajirao University of Baroda.

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Abstract

Digital holography is an emerging field of new paradigm in general imaging applications. With the advancement of this technique into Digital holographic microscopy (DHM), many biomedical applications have emerged and helped in various fields. Because there are many fatal genetic disorders that causes diverse morphological abnormalities in erythrocytes (RBCs), it is critical to investigate the three-dimensional (3-D) structure of RBCs as a tool for detecting the disease. The use of self-referencing common path 3-D digital holographic microscope using lateral shearing interferometer for the purpose of imaging and analysis of the static and dynamic parameters of disease affected RBCs is described in this study. This information can be utilised to calculate thickness, volume, surface area to volume ratio, eccentricity, sphericity, membrane fluctuation amplitude, and frequency, among other morphological properties. These parameters of the diseased and the healthy cells were analysed and a comparative study was done to differentiate and characterize the cells. Using Machine Learning, Logistic Regression, SVM and other technical algorithms, a unique program was synthesized to test a random blood sample whether it is healthy or a diseased sample.

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CHAPTER 1
Introduction

Observation is the key to making new discoveries. When it comes to observing the materials, there is a limitation to the size of the objects which a human eye can see and distinguish. Therefore, an instrument which can magnify such small objects and bring them to a scale that can be suitable for a human eye to see and distinguish is compulsory. A microscope is such an instrument which is used since ages to study the structure and properties of the small-scale materials around us which are beyond the capabilities of a human eye to see and distinguish. A microscope is basically an apparatus similar to the design of a telescope which comprises of a lens or a combination of lenses. It can magnify a small object to tens or hundreds of times which helps in visualizing various objects ranging from biological to mineral samples which has opened up a whole new dimension of science.

For most of the objects that are visualised using microscopy, the microscope uses the natural light emitted by the object to form an image of that particular object. Objects with different colours will absorb different wavelengths of light and will also reflect a spectrum of wavelengths which we see as the colour of the object and this is mostly true for mineral samples. But when it comes to the biological samples, there is a catch, while most of the biological samples absorb and scatter a few wavelengths of light, there are samples that are transparent which absorb very few wavelengths of light in the visible regime and will reflect most of the light. When this case arises, it is difficult to observe the sample through a conventional microscope. There are many methods which are developed which can help enhance the view of object such as staining the sample with dyes, etc. In such methods, the sample is stained with a contrasting agent. But such methods involve manipulation with the sample which is not suitable for living cell samples as the cells may die or may change their behaviour or properties coming in contact with the contrasting agent. Moreover, the conventional microscopes face a hurdle when it comes to visualize the properties of a sample in 3-D as it can only provide object information at a certain 2-D plane.

In order to overcome this hurdle, a technique which can capture the amplitude contrast as well as the phase of the object was needed. A scientist named Gabor came with such a technique which he named “Holography” which means “whole writing”. After this invention, many advancements occurred in this field and a new technique which involved the numeric reconstruction of hologram came into play which is called “Digital Holography (DH)” which

allowed the holograms to be captured by a CCD camera and read the image as a matrix which further helps to calculate the phase reconstruction of the captured hologram.

With the invent of this technique, the conventional microscopy was replaced by the "Digital Holographic Microscopy (DHM)". Using this Digital Holographic Microscopy (DHM), one could construct and analyse images of minor objects which are even transparent in nature. The parameters such as thickness, volume, surface area to volume ratio, eccentricity, sphericity, membrane fluctuation amplitude, frequency, etc of small cellular structures can be easily interpreted. As holographic imaging is directly related to the phase profile of the object, the Holographic Microscopy is also referred to as Phase Microscopy.

Using Quantitative Phase Microscopy in medical purposes, the morphological features of RBCs altered by diseases are investigated. Because no external substances are employed in this procedure, the cells native state is preserved. This method generates a three-dimensional thickness profile of the cells as well as thickness fluctuations over time. The optical thickness depends upon the refractive index of the object as well as on its actual physical dimensions. Various physical and mechanical parameters of RBCs can be derived using this information. Comparing all those parameters obtained, a general classification of the cells can be carried out as will be discussed later in this study.

CHAPTER 2
Holography

Holography was invented in 1948 by Dennis Gabor in an effort to improve the resolution of the electron microscope, where the correction of electron lens aberrations posed increasing technical difficulty. Instead of attempting to perfect the electron imaging lens, Gabor dispensed it altogether and realized that the diffraction pattern of the electron beam contained complete information regarding the amplitude and phase of the electron wave. The record of the electron wave diffraction is then used to optically synthesize the object field. This allows the use of visible light for image formation, and was much easier and more developed compared to electron optics. He named the new imaging principle holography, for its ability to record the whole optical field.[1]

The holography principle was immediately applied to recording and imaging by visible light. But it had to wait for two critical inventions before its full potential was to be realized. One was the powerful coherent source of light in lasers to provide high quality interference contrast. The other, due to Leith and Upatniek's, was off-axis illumination with a separate reference wave, thus eliminating the problem of the zero-order and twin images of the Gabor's on-axis configuration. Soon many new techniques and applications of holography began to develop.[1]

It was also realised early on that the use of non-planar structures of the reference beam can lead to various manipulations of the holographic output, ranging from image magnification to more complex image processing, such as encryption, pattern recognition, associative memory, and neural networks. Instead of photographic plates or films, real-time holography is possible with photorefractive and other nonlinear optics materials. In fact, we now understand much of nonlinear optics as generalizations of the holography principle, including phase conjugation, four-wave mixing, spectral hole burning, and photon echo. The holography process is being developed for other regions of the electromagnetic spectrum, especially X-ray holography, with its prospect of atomic resolution, as well as for micro holography of living organisms. [1]

2.1. Basics of Holography

A hologram is usually recorded on a photographic plate or a flat piece of film, but produces a three-dimensional image. In addition, making a hologram does not involve recording an image in the conventional sense. To resolve this apparent paradox and understand how holography works, we have to start from first principles. [2]

In conventional imaging techniques, such as photography, what is recorded is merely the intensity distribution in the original scene. As a result, all information about the optical paths to different parts of the scene is lost. [2]

The unique characteristic of holography is the idea of recording both the phase and the amplitude of the light waves from an object. Since all detecting materials respond only to the intensity in the image, it is necessary to convert the phase information into variations of intensity. Holography does this by using coherent illumination and introducing, as shown in the figure below, a reference beam derived from the same source. The photographic film records the interference pattern produced by this reference beam and the light waves scattered by the object. [2]

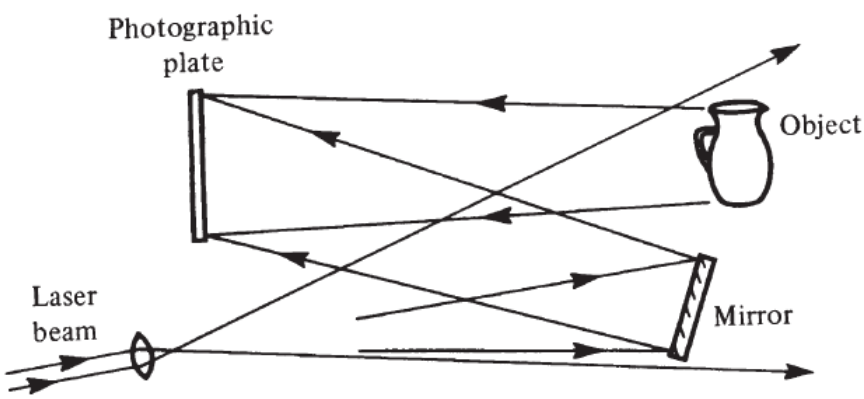


Fig (2.1): Hologram construction

Since the intensity at any point in this interference pattern also depends on the phase of the object wave, the resulting recording (the hologram) contains information on the phase as well as the amplitude of the object wave. If the hologram is illuminated once again with the original

reference wave, as shown in the second figure above, it reconstructs the original object wave.[2]

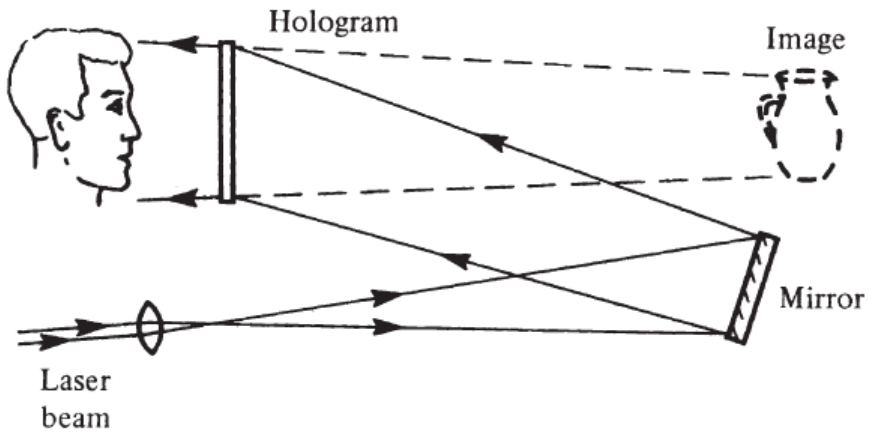


Fig (2.2): Image Reconstruction

An observer looking through the hologram sees a perfect three-dimensional image. This image exhibits, as shown in figures above, all the effects of perspective, and depth of focus when photographed, that characterized the original object. [2]

There are mainly two ways in which a hologram is captured I.e., The In-line setup and the off-axis setup. In the in-line setup, the object beam as well as the reference beam both travel parallel to each other whereas in an off-axis setup, the object beam and the reference beam are incident forming a small angle with each other. The main difference between this two setups is that while reconstruction, one can see multiple images superimposing in the in-line setup because the real image as well as the virtual image are formed on the same axis whereas in the off-axis setup, this is not a problem as the real and virtual images are formed on either sides of the principle axis at a certain angle. The figures describing the two ways are as shown below.

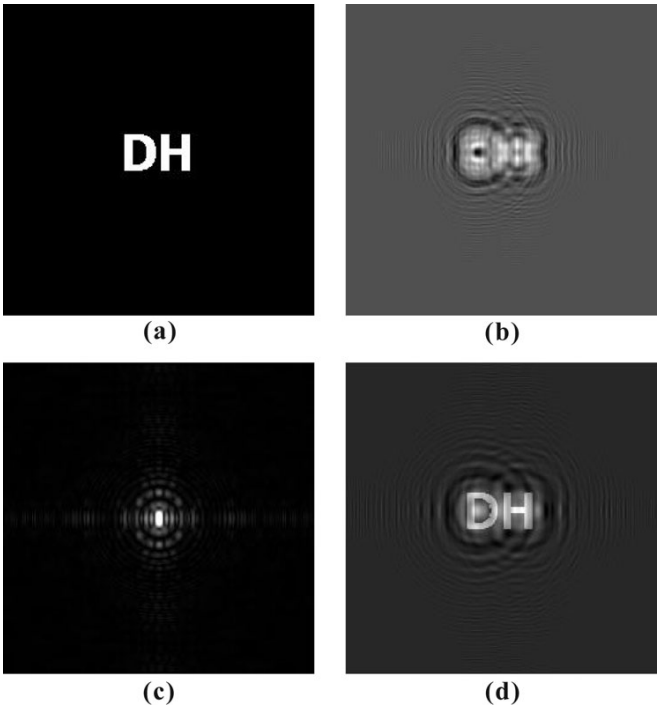


Fig (2.3): (a) Input image(original object), (b) on-axis hologram,

(c) spectrum of (b), and (d) reconstructed image.

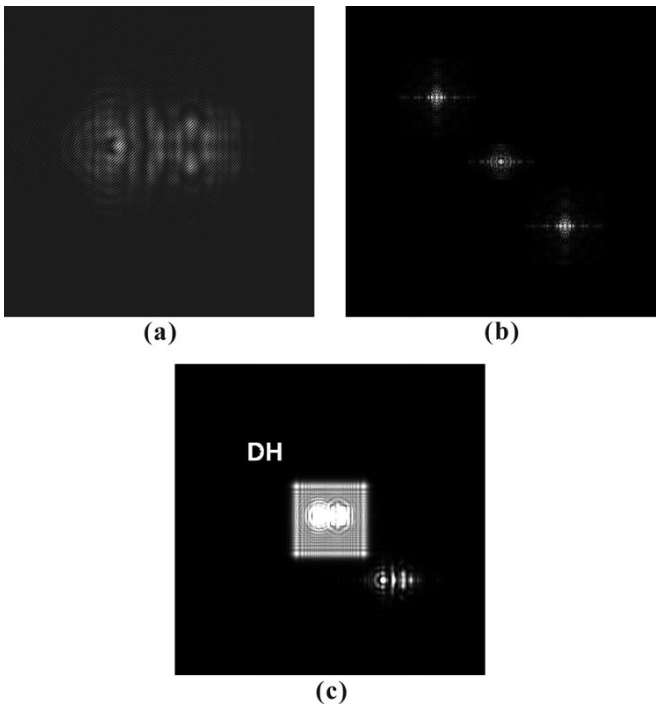


Fig (2.4): (a) Off-axis hologram, (b) spectrum of (a), and

(C) reconstructed image.

2.2. Digital Holography

For many application areas, real-time operation is of critical importance but is difficult with conventional holography. Even photorefractive and other nonlinear optics systems require substantial equipment and technical care to implement them, and have seen only limited practical applications.[1]

Digital holography replaces physical and chemical recording processes with electronic ones, and the optical reconstruction process with numerical computation. The propagation of optical fields is completely and accurately described by diffraction theory, and in 1967, Goodman and Lawrence demonstrated the feasibility of numerical reconstruction of an image from a Fourier hologram detected by a vidicon camera. Schnarr's and Jueptner, in 1994, were the first to use a CCD camera directly connected to a computer as the input, and compute the image in a Fresnel holography setup. [1]

In what is now called digital holography (DH), holographic interference is produced by optical processes in real space, while reconstruction is by numerical computation. Conversely, in computer generated holography (CGH), the hologram can be produced by numerical computation inside a computer, followed by printing or other outputs to real space. Reconstruction is then carried out by optical means. CGH has many interesting properties and applications, such as the ability to arbitrarily prescribe desired amplitude and phase properties of the output optical field starting from fictitious objects. [1]

By direct electronic recording of holographic interference, and because of the increasing speed of holographic computation, real time holographic imaging is now possible, and more importantly, the complete and accurate representation of the optical fields as an array of complex numbers allow many imaging and processing capabilities that are difficult or infeasible in real-space holography. Various useful and special techniques have been developed to enhance the capabilities and extent the range of applications.[1]

In digital holographic microscopy, a single hologram is used to numerically focus on the holographic image at any distance. Direct access to the phase information leads to quantitative phase microscopy with nanometre sensitivity of transparent or reflective phase objects, and allows further manipulations such as aberration correction. Multiwavelength optical phase unwrapping is a fast and robust method for removing 2π-discontinuities compared to software algorithm-based methods. [1]

A significant constraint of digital holography is the pixel count and resolution of the imaging devices. Suppression of the zero order and twin images by phase shifting digital holography allows efficient use of the pixel arrays. Digital Gabor holography, without separate reference beams, is useful for particle imaging applications by providing 4-D space-time records of particle fields. [1]

Digital holography naturally evolved from the effort to utilize electronic imaging in interferometry, such as in electronic speckle pattern interferometry (ESPI). Metrology of deformations and vibrations is a major application area of digital holography. Optical processing, such as pattern recognition and encryption, by digital holography also offers new capabilities. [1]

In this study, the holograms are constructed optically and are reconstructed numerically using scalar diffraction theory in MATLAB. There are various approaches for this purpose but the most used and the most effective and fastest approach is the Fourier analysis. The Fast Fourier Transform, which is an advanced version of Fourier transform allows the calculations to take place within a very short interval of time such that real time calculations are possible and hence opening a gateway towards real time imaging. Fourier fringe analysis of the recorded holograms yields the object wavefront's complex amplitude, from which phase information can be derived, resulting in the object's thickness distribution.

2.3. Digital Holographic Microscopy (DHM)

Digital holographic microscopy (DHM) is a technique which enables the numerical reconstruction of recorded holograms with a computer. In DHM, amplitude and phase information of a target object is recorded on a charge-coupled device (CCD) camera as an interference pattern, which is called hologram, through an objective lens. Then, similar to an optical holography, the diffraction of wavefronts from the recoded hologram is computed and numerically propagated along the reconstruction distance to any chosen observation plane with a computer system. Thus, based on this numerical knowledge of the propagated complex wavefront, the amplitude and the phase of a target object can be computed, which can lead to micrometer- or nanometer-scale quantitative measurements of the target object.[3]

When investigating, DHM obtains the depth information of a specimen and resolves phase differences corresponding to depth differences as price as several nanometres. Thus, it would be well suited for detection of nanometer- or micrometer-scale defects on transparent objects. Furthermore, since the phase information provided by DHM allows us to visualize the detailed topological shapes of those defects, DHM attracts the attentions of various information technology industries. [3]

In general, DHM can be implemented with an interferometer system. There are two kinds of interferometer systems which are two-beam and single-beam interferometers, which include the Michelson and the Mach-Zehnder interferometers and shearing interferometer, respectively. Here, the recorded hologram by DHM is usually composed of three components which are DC bias and real and virtual images, in which DC bias and virtual image are undesired terms in the numerical reconstruction process. Thus, these terms need to be eliminated because their elimination results in an enhancement of image quality.[3]

CHAPTER 3
Experimental Setup and Preparations

A basic digital holographic microscopy (DHM) setup consists of an illumination source, an interferometer, a digitizing camera, and a computer with necessary programs. Most often a laser is used for illumination with the necessary coherence to produce interference. All different types of lasers have been used, from uniquitous He-Ne lasers and diode lasers, to diode-pumped and doubled YAG lasers (often referred to simply as solid-state laser), argon lasers, as well as tunable dye lasers and Ti-sapphire femtosecond lasers. For multiwavelength techniques, two or more different lasers can be coupled into the interferometer, or a tunable laser can be employed. There are also low-coherence techniques for the purpose of reducing speckle and spurious interference noise, or generating contour or tomographic images. A short-pulse laser can be used, or a tunable laser can be turned into a broadband source by removing the tuning element. Even an LED typically has 10 micro meter or so coherence length, which can be sufficient for holographic microscopy. [1]

The current study's setup is based on self-referencing common path geometry. In self-referencing geometry, the reference wavefront is a section of the object that is not modified by object information. The simplest approach to use a holographic microscope in self-referencing mode is to use a glass plate to create two laterally shared versions of the same wavefront. The pictures from the front and back surfaces of the glass plate do not overlap when the degree of shearing is greater than the size of the magnified objects, generating holograms rather than shearograms. These holograms are reconstructed in the same way that conventional two-beam digital holograms are.

Figure 3.1 depicts a schematic of the experimental setup, whereas Figure 3.2 depicts a photograph of the microscope. Using a mirror at 45^0 to the angle of incidence of the laser beam, a He-Ne laser beam ($\lambda = 611.9\,nm$, random-linearly polarised) is used to trans-illuminate the object (placed on a translation stage). A microscope objective lens (34x, NA=0.65) magnifies the object. Two laterally sheared copies of the object wavefront were created using a fused silica glass plate with a thickness of 7mm. To record the holograms, a CCD array (8-bit monochrome, 1024x768 pixels, 4.65 m pixel pitch) was mounted at the picture plane of the objective lens.

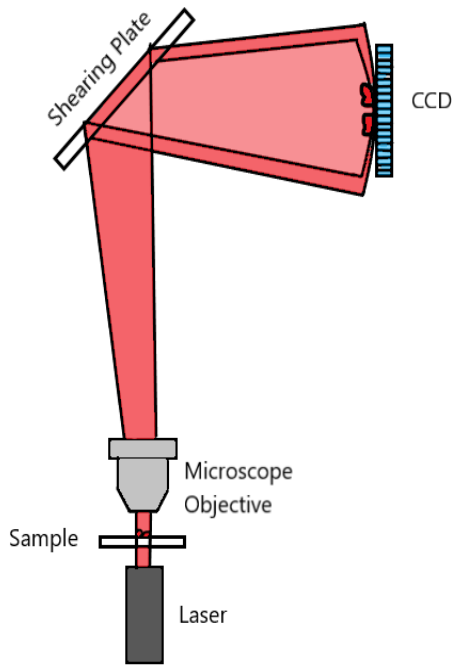


Fig (3.1): Schematic diagram of microscope employing lateral shearing interferometry.

The expanding source beam, coupled with the object information, strikes the front surface of the glass slab, where 4% of the incident intensity is reflected, while the remaining 96 percent is transmitted and reflected back from the back surface. As the wavefronts reflected from the front and rear surfaces traverse different optical path lengths, they will be spatially and axially separated. These two wavefronts overlap and generate a hologram, which is captured using a digital sensor.

The true strength of the lateral shearing interferometer is that both the object and reference beams travel along the same path and pass through the same set of optical components. When compared to a digital holographic microscope with a Mach-Zehnder interferometer arrangement, where the object and reference beam travel along different routes and encounter various sets of optical elements, this LSI arrangement results in great temporal stability. This device gives sub-nanometer temporal stability for a long period of time over several minutes, making it perfect for measuring thickness changes ranging from a few nanometers to a few tens of nanometers, as seen in red blood cells.

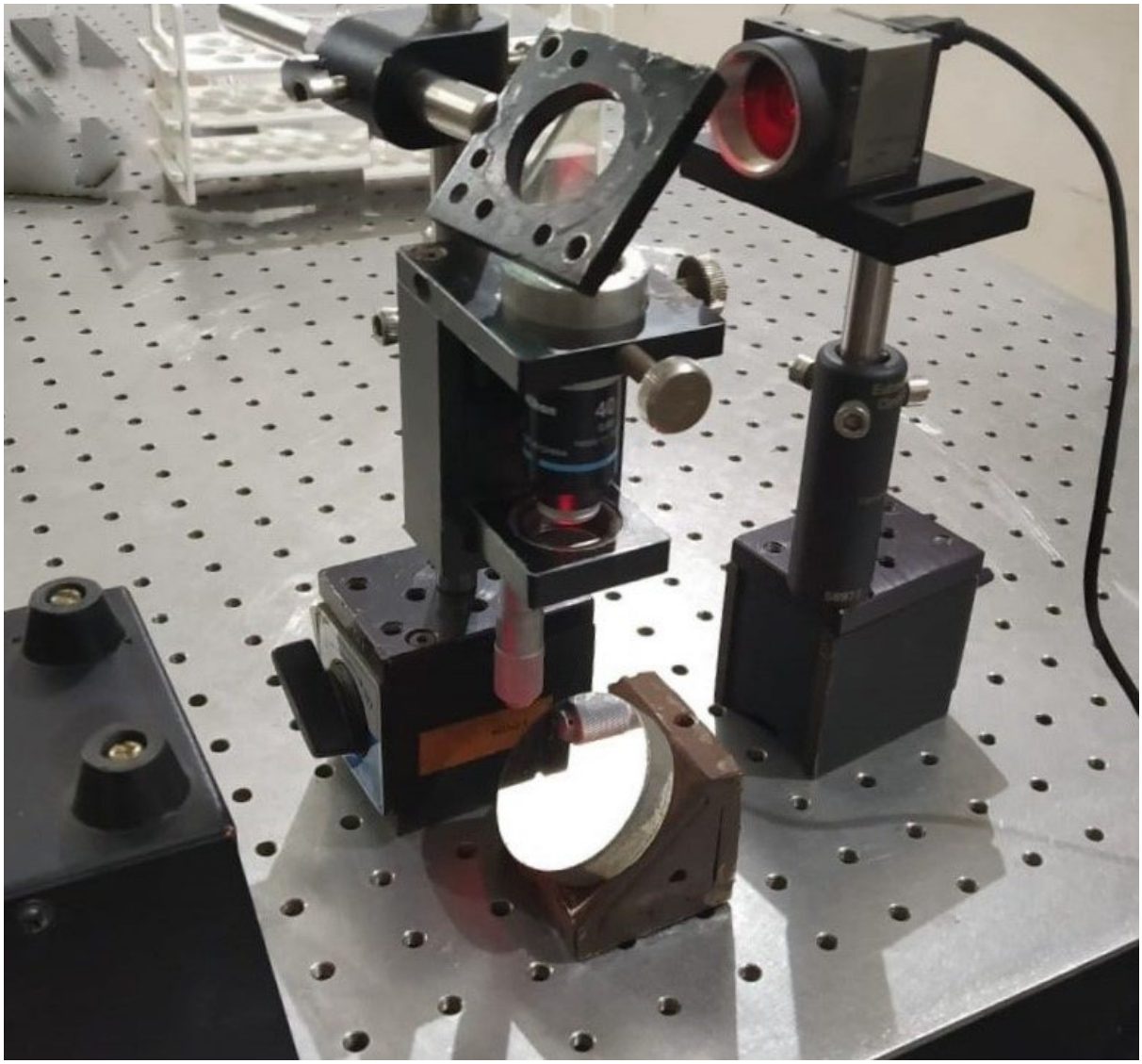


Fig (3.2): Photograph of microscope employed in this experiment

3.1. Focusing

For the extraction of accurate phase of microscopic samples, focusing of the sample is crucial. A small defocus leads to a significant phase error. In two beam interference microscope, sample can be brought to focus by blocking one of the beams (reference) and looking for the highest contrast image. The plane of highest contrast can be considered as the plane of focus. However, two beams always overlap in shearing interference microscope. It is difficult to first isolate and then look for the sharp bright field image. We propose a method to find the focused hologram out of a number of recorded holograms. [5]

As phase information of the object is encoded in terms of the modulated fringes, the maximum modulation of fringes correspond to the phase of the object. Therefore, a well focused hologram is one where in fringe modulation are maximum. Modulation in fringes can be calculated in terms of the total gradient of the image. Which is called as the total gradient(TV) of the image. Total variation is the gradient of the image summed over all the pixels. Therefore, for the best focused hologram, total variation is maximum. Stack of holograms were recorded across the focus plane and one with the maximum TV was labelled as focused hologram. [5]

3.2. Recording A Hologram

For recording a hologram using the proposed system, region of interest was chosen very carefully, taking into account the object location, thickness and refractive index of the shear plate and the angle of incidence at the shear plate. In conventional systems, the duplicate image of the object could not be removed due to small lateral shear distance. This is because more than half of the object beam overlaps with itself. This problem was taken care by maintaining the lateral shearing distance less than or equal to half of the FOV in the proposed setup. [5]

A total of 6 samples were collected: 2 for Thalassemia beta trait, 2 for sickle cell disease and 2 from healthy volunteers. A total of 1432 RBCs were extracted from holograms of these samples.

CHAPTER 4
Imaging of human red blood cells and calculating parameters

In certain cases, diseased RBCs can be differentiated from healthy ones on the basis of phase. Phase of RBCs present in human blood sample is imaged using the proposed setup. The unstained blood smear was prepared in the lab and the periphery of the blood smear was scanned using a 40x M.O. (0.65 N.A. for wide FOV). The slide was kept on the sample holder of the microscope and a series of holograms were recorded for a time span of 20s(@ 15 frames per second) for analysis of time variation of thickness profile.

The physical parameters of the cells were determined using a single hologram from this time series. The slide was translated after imaging one field of view to obtain images from a different field of view. A single thin smear often yielded 40-60 field of views. Within 10 minutes of preparing the thin smears, the holographic imaging of the cells was finished.

Fig. 4.1a shows portion of the recorded hologram in the case of healthy red blood cells. Inset of the figure shows the region inside the rectangle showing modulation of interference fringes. Reconstruction is achieved using angular spectrum propagation integral. Fig. 5.1b shows the reconstructed intensity at the image plane. A reference hologram of the medium (blood plasma) around the cells was also acquired and reconstructed. Subtract the phase acquired from the reference hologram from the phase obtained from the object hologram. Because the phase due to aberrations remained consistent between the two exposures, this phase subtraction nullifies most of the undesired phases (mostly due to aberrations). Fig. 4.1c shows the quantitative phase contrast image obtained after phase subtraction. The cells' thickness distribution is calculated using a constant average refractive index of 1.42 for red blood cells and 1.34 for the surrounding medium (blood plasma). Thickness distribution of healthy red blood cells is shown in Fig. 4.1d. The doughnut shape of the red blood cells is quite evident in Fig. 4.1d. Also notable is that the size, shape and thickness of the cells are similar.

Similar holograms were captured for the diseased cells such as the sickle cells and also the cells affected by thalassemia. Their phase reconstruction was carried out by the same procedure we discussed above and also the calculations for thickness distribution was carried out.

The cell morphology is determined by the thickness distribution acquired from phase profiles. This profile yielded several cell parameters depending on its shape. To establish the position of the cells, the cell thickness profile is thresholded with the background thickness profile (area

with no cell in the field of vision) before extracting the morphological parameters. Cells in the field of vision are automatically labeled after thresholding (Fig. 4.3a). The thickness distribution inside a cell region is utilised to derive morphological features specific to that cell once it has been identified and labelled. Therefore, we calculated the physical and mechanical parameters pertaining to the cells which were clearly visible without clustering with the surrounding cells to get better results.

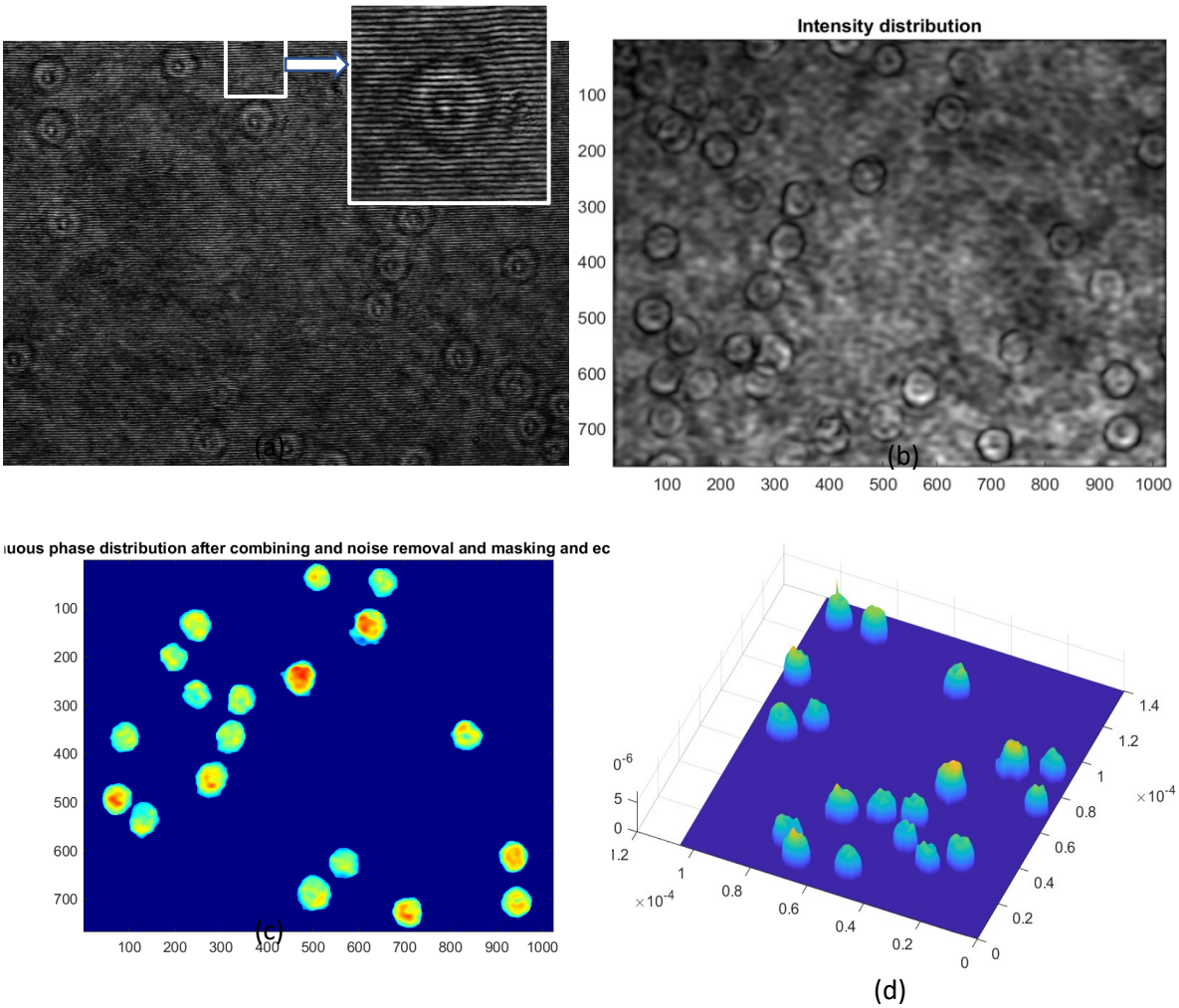


Fig (4.1): (a)Hologram of healthy red blood cells. (b)Reconstructed intensity at the image. plane. (c)Quantitative phase image. (d) Thickness profile

CHAPTER 5
MACHINE LEARNING FOR CELL CHARACTERIZATION

Machine learning (ML) is a part of artificial intelligence (AI) that allows software applications to improve their prediction accuracy without being exclusively designed to do so. Machine learning algorithms estimate new output values using historical data as input.

Machine learning is a large field that takes a completely different approach to modelling, relying on very little human knowledge. The most common subdiscipline is supervised learning which assumes that the relationship between the measured and predicted variables is straightforward in some way with features like smoothness, sparseness, and invariance.

Supervised algorithms take feature vectors as inputs and output predictions. They differ from the unsupervised learning algorithms in the fact that they have to be provided with labelled data or output classes along with the input variables.

In this study, we have used Support Vector Machine as the supervised learning algorithm to classify between diseased and healthy cells.

5.1. SUPPORT VECTOR MACHINE

Support Vector Machine, or SVM for short, is a linear model for classification and regression issues. It can solve both linear and nonlinear problems and is useful for a wide range of applications. The goal of the SVM algorithm is to find a hyperplane in an N-dimensional space that categorizes data points clearly. The hyperplane's size is determined by the number of features.

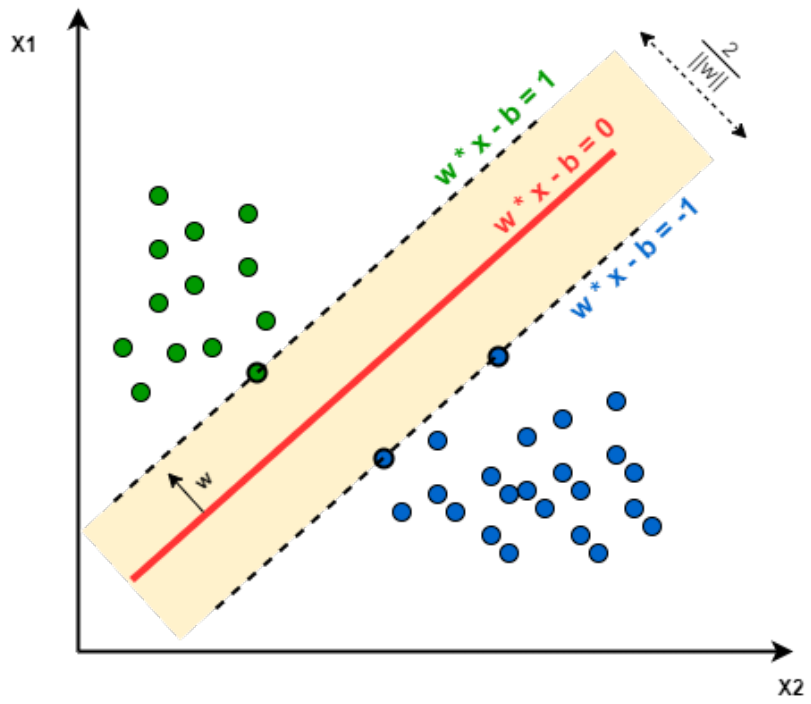


Fig (5.1) : SVM with a 1-d hyperplane

If there are only two input characteristics, the hyperplane is just a line. When there are three input features, the hyperplane becomes a two-dimensional plane. When the number of features exceeds three, it becomes impossible to imagine. When the data are not linearly separable, SVM overcomes this by utilising a kernel to create a new variable. The new variable y is formed as a function of distance from the origin in this scenario. The SVM kernel is a function that converts non separable problems into separable problems by transforming low-dimensional input space into higher-dimensional space. It is most beneficial in cases with non-linear separation. Simply said, the kernel performs some fairly sophisticated data transformations before determining how to split the data using the labels or outputs specified.

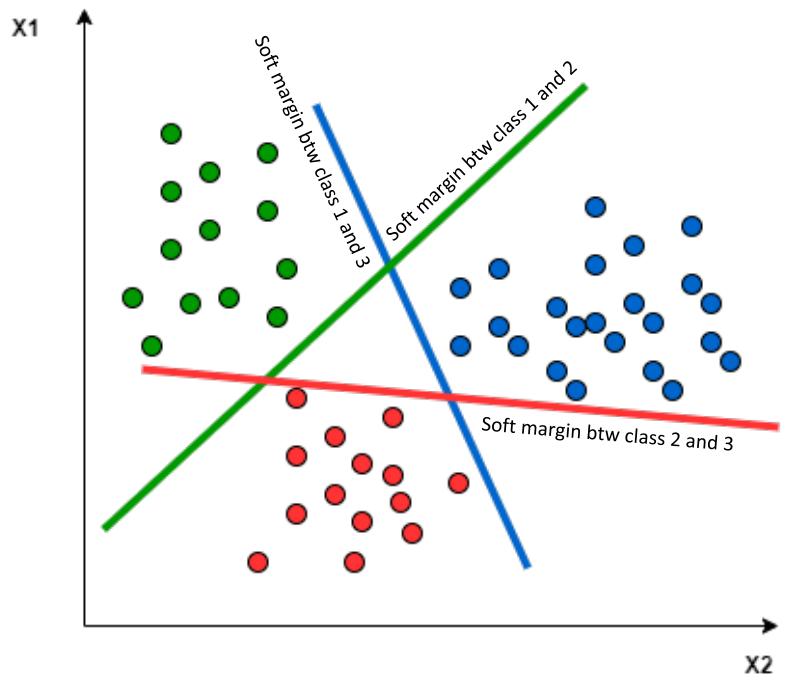
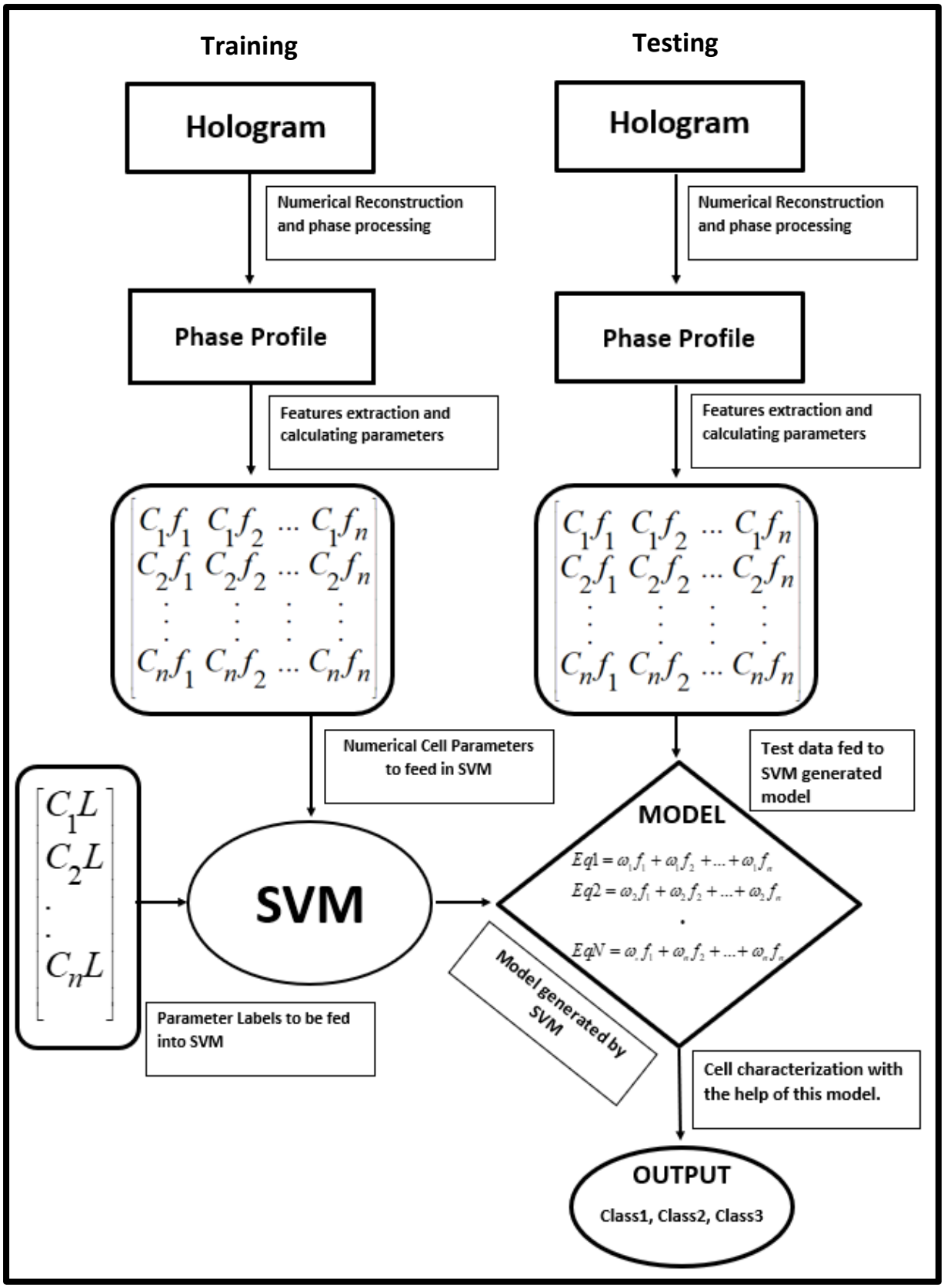


Fig (5.2): SVM with 3-D hyperplane

As we can see in the figure above, if there are three different data sets with different characteristics, the SVM algorithm will try and create a hyperplane for every pair of two data sets. Therefore, the SVM will try and hence will generate soft margin or a hyperplane between data set one and data set two, data set two and data set three as well as a hyperplane for data set one and data set three. A model will be generated combining the equations of these three hyperplanes. When new data is imported inside this newly generated model, the data is classified.



Fig(5.3): Flowchart of data classification using SVM

In our study, we calculated and merged all the different parameters of all the cells containing the diseased cells (i.e., sickle and thalassemia) as well as the healthy cells. After that we divided our data into two parts, i.e., the data needed to train the algorithm and the data that is needed to test the outcome.

All the training numerical data was fed into the SVM algorithm along with their labels (i.e., 0 for healthy cells, 1 for sickle cell disease and 2 for thalassemia). After being fed with all these data, the SVM algorithm will differentiate all these data based on their different parameters and will create a model to identify the new test samples. The machine will train itself by using these data provided and increase its accuracy. The more data provided, the better results we get.

The model generated by the SVM algorithm is basically a group of equations of hyperplanes generated by the SVM algorithm with the help of the training data that is fed to it. This group of hyperplanes provide a maximum soft margin for the upcoming data to be classified. In this study, the data contained three categories of cells and cells from all these categories were fed to the algorithm. Hence, the algorithm will carry out the classification between all these three types of data one by one with the help of the labels provided. A hyperplane was generated for each pair of categories of cells. (Hyperplane 1: between 0 and 1, Hyperplane 2: between 1 and 2, Hyperplane 3: between 1 and 3). Therefore, the model generated in this study, will have three equations of hyperplane.

Now, the data which was reserved earlier for the testing purpose was fed to this model in SVM. The newly fed data contains the label as well, but the algorithm will hide this label and will categorise the cells based on the model generated with three hyperplane equations. After the categorisation is completed, the algorithm will unhide the labels and will compare the results generated with the original test data. Hence, the degree of accuracy of the model generated and the data categorisation is obtained.

CHAPTER 6
RESULTS AND CONCLUSION

EXTRACTED CELL FEATURES:

With the use of thickness profile, we obtained after reconstructing the hologram, we calculated various physical and mechanical parameters as shown below:

Physical	Mechanical
Diameter	Optical Path Length
Thickness	Volume change
Area	Frequency fluctuations
Volume	Lateral oscillations
Eccentricity	Membrane stiffness
Sphericity	Dynamic light scattering
Skewness	
Kurtosis	

Probability distribution for mean thickness for the three classes are shown in Fig. 6.1. Two-sample t-test confirmed that the extracted features come from independent random samples having normal distribution.

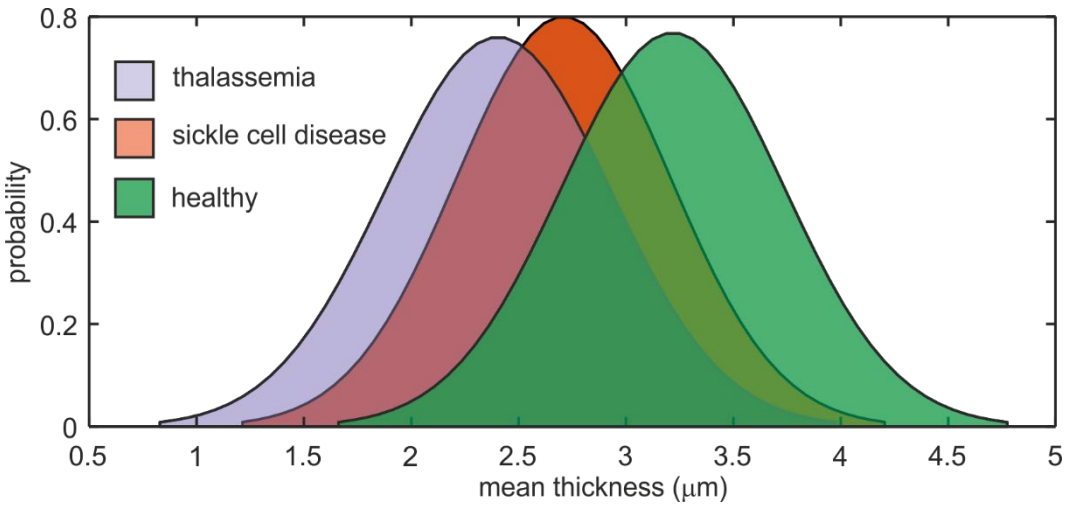


Fig 6.1: Thickness distribution for the three classes of red blood cells

Cell volume also showed a similar trend with appreciable difference between the distributions of different classes as shown in Fig 6.2. Figures 6.1 and 6.2 shows the biophysical features of the cells. From a series of holograms, biomechanical cell features can also be extracted Fig. 6.3 shows the distribution of volume fluctuation for the three classes of red blood cells under investigation. This plot also shows significant difference between the classes.

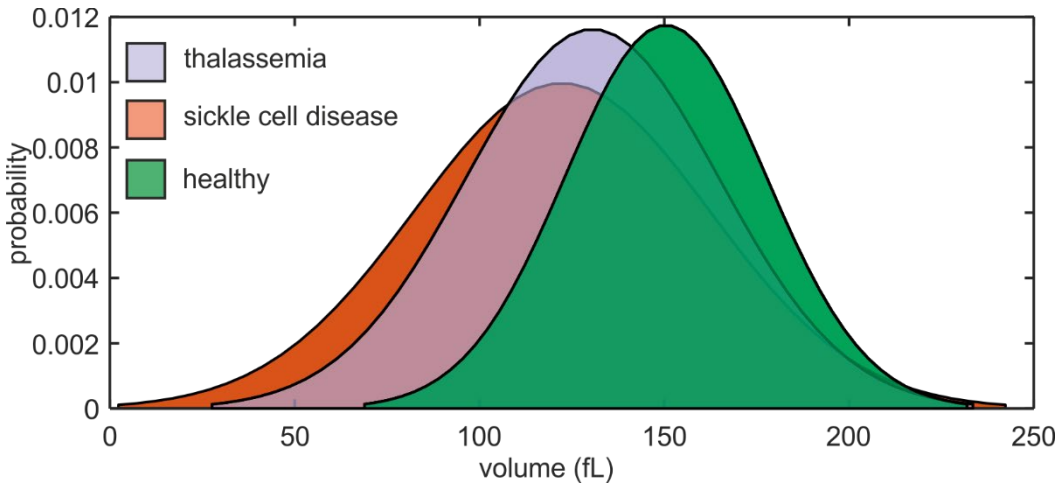


Fig 6.2: Volume distribution for the three classes of red blood cells

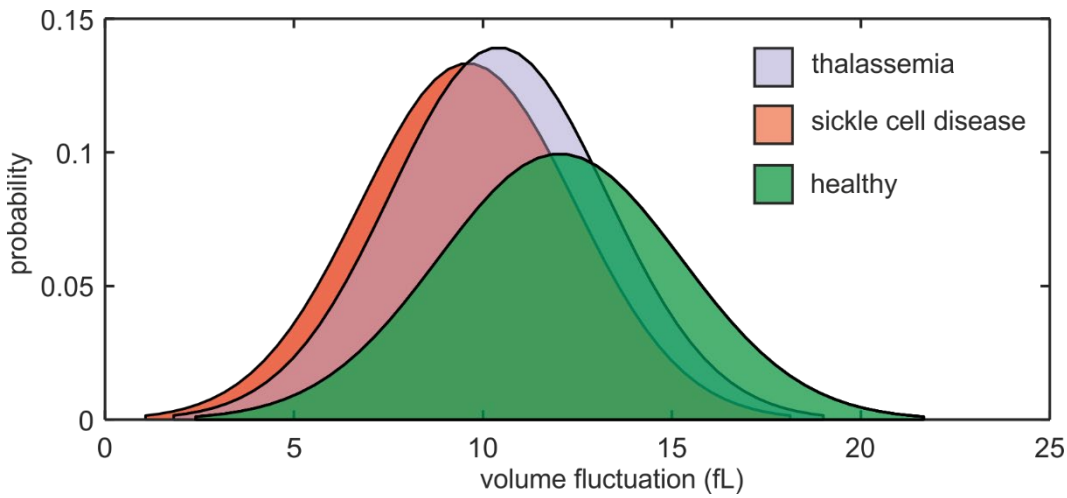


Fig 6.3: Distribution of Volume fluctuation for the three classes of red blood cells

CLASSIFICATION ACCURACY:

The classification of the cells as health and diseased was carried out using the trained SVM model. Fig. 6.4, shows the confusion matrix for the classification accuracy of the trained model. The sensitivity for healthy red blood cells was 92.4% (means they were correctly classified as heathy red blood cells) and the true positive rate (sensitivity) for cells from diseased class (thalassemia + sickle cell) was 85.6%.

The overall accuracy of the classification of the cells between the healthy and the diseased was found to be 89%.

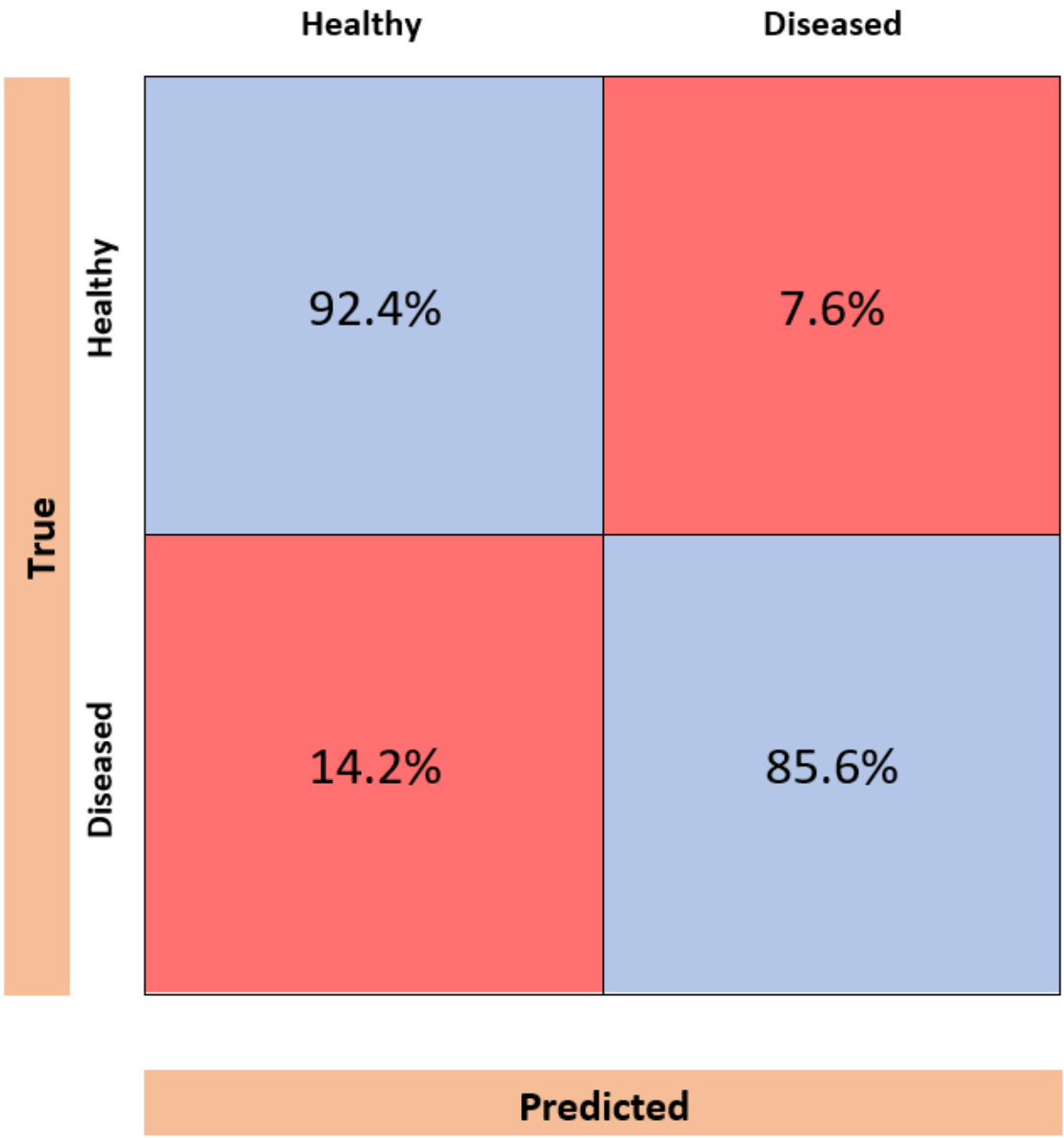


Fig. 6.4: Classification accuracy for healthy and diseases class

The Receiver operating characteristics (ROC) curve is shown in Fig. 6.5. This curve is a plot between True Positive rate vs False positive rate. This curve represents the optimum threshold obtained by the SVM for the classification of the cells for the model to work. The curve in this figure is obtained between two classes, i.e., the healthy cell and the diseased cells. The operating point at which the model operates lies on this ROC curve. Here, the OP is (0.14,0.92).

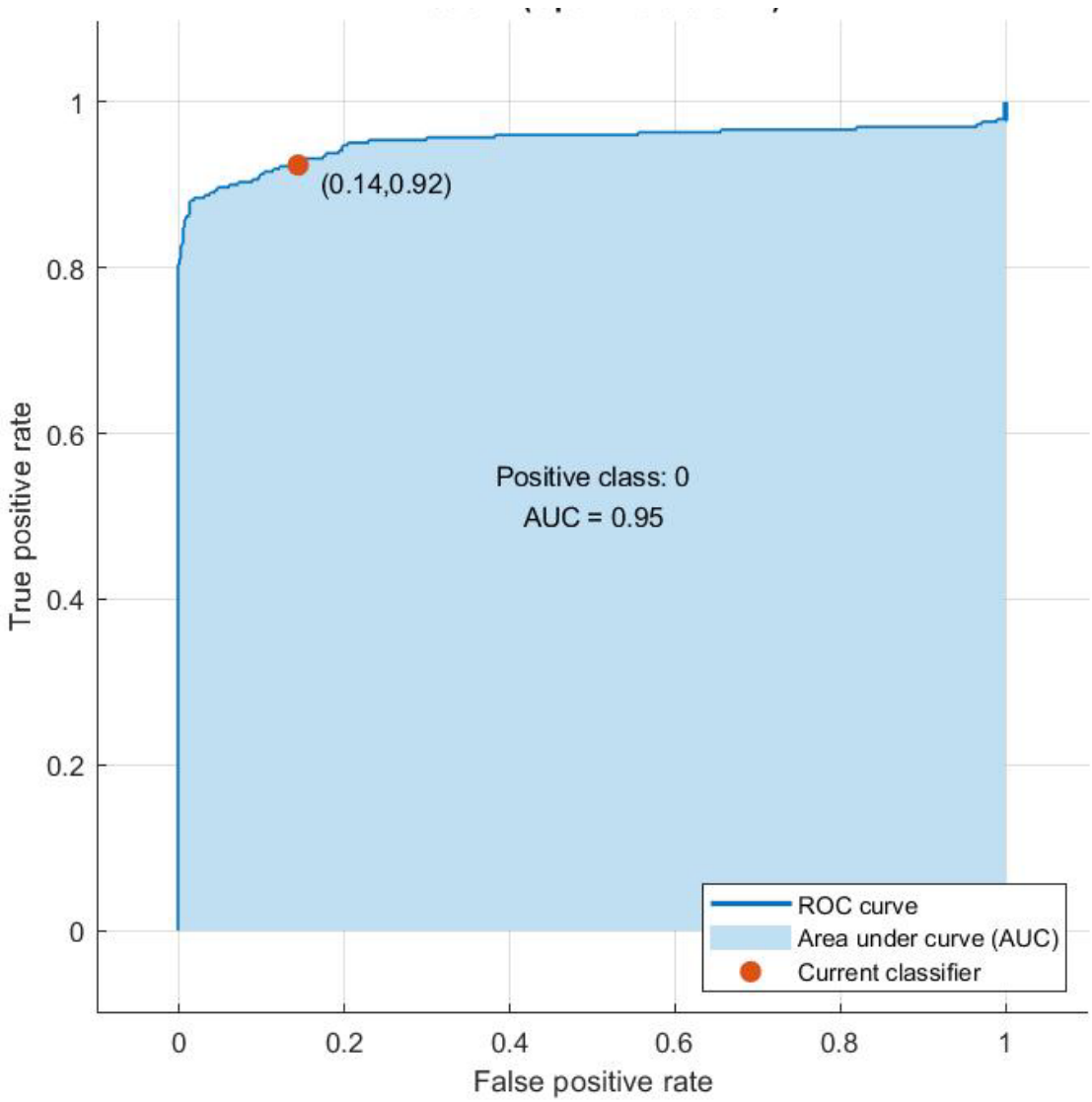


Fig. 6.5: Receiver Operating Characteristics

With the help of SVM machine learning algorithm and the trained model, we got the result of the classification of the cells in the form of confusing matrix. The confusion matrix as shown in Fig. 6.5 shows the multi-class classification ability of the trained model. It depicts accuracy of the technique to classify the red blood cells into healthy, thalassemia and sickle cell classes. The elements of matrix provide the percentages of true positive rate, true negative rate, false positive rate as well as false negative rate.

The true positive rate and the true negative rate gives the percentage of truly identified cells whereas the false positive rate and the false negative rate gives the percentage of the cells which were falsely classified.

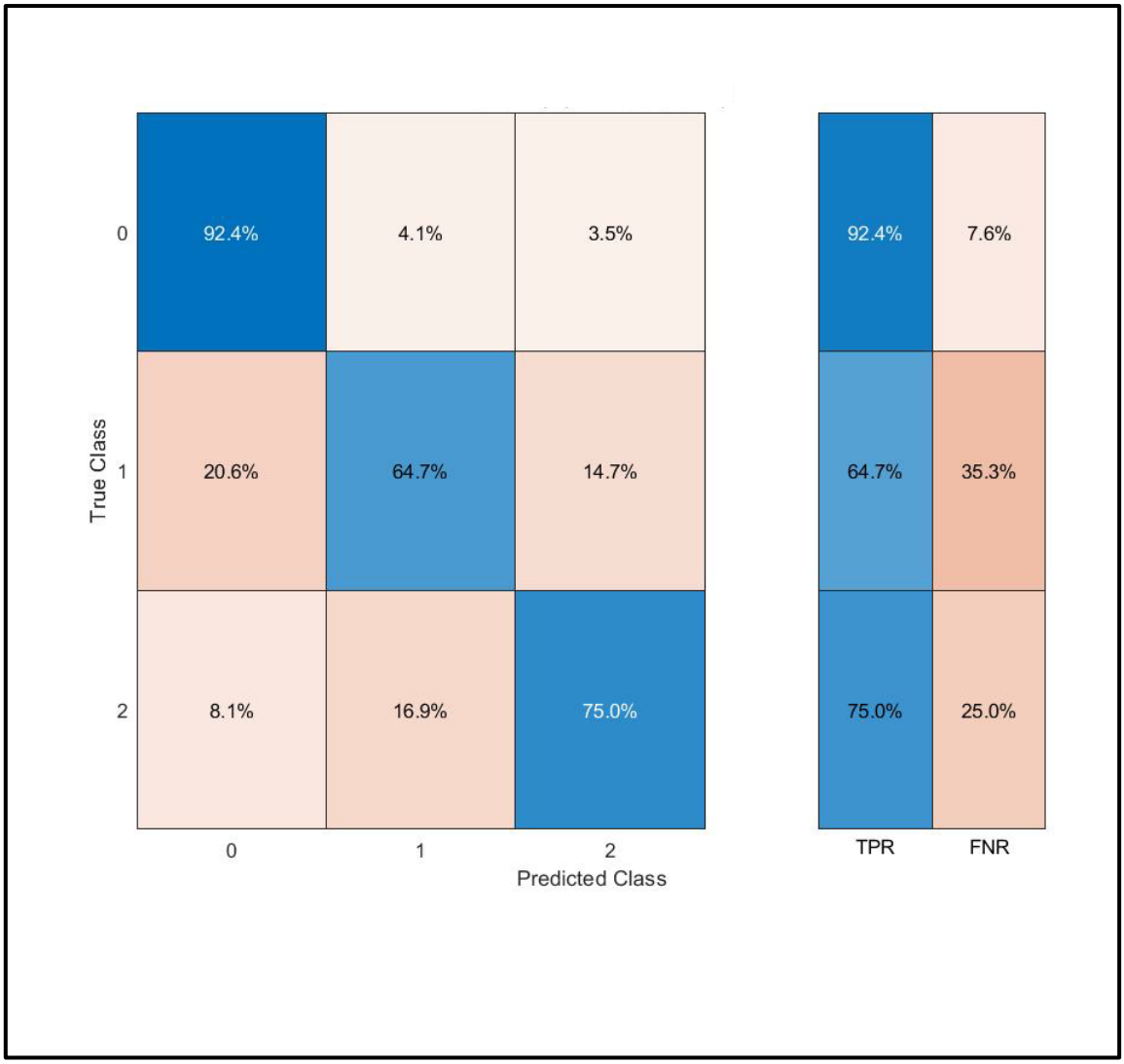


Fig 6.6: Confusion matrix for the trained SVM model for multi-class classification (classification of the data into healthy, thalassemia and sickle cell)

The percentage of accuracy for each label of cells is shown in the above matrix. The matrix also shows the percentage of false or incorrect predictions made by the algorithm. Here, the data is shown as True class data vs Predicted class data. The overall accuracy of the classification of the cells between three classes, i.e., the healthy, the sickle and the cells influenced by thalassemia was found to be 77.3%.

CONCLUSION:

In this study, we did stain-free imaging of healthy and diseased red blood cells using Lateral Shearing Interferometer. Bio-physical and bio-mechanical parameters were extracted for all the cells. SVM classified the cells into healthy and diseased with an accuracy of 89%. It also classified the cells into healthy, sickle and thalassemia with an overall accuracy of 77%. Since all the RBCs are not affected with sickle-cell disease or thalassemia, making group of cells from each disease and then feeding to the algorithm can improve the classification accuracy. Also, other algorithms like neural network can be used to classify the cells with better accuracy.

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